

Installation & Deployment Guide — v1.0

ຄູ່ມືຕິດຕັ້ງ ແລະ Deploy ລະບົບເທິງ Google Apps Script + Google Sheets + Google Drive. ຮ້າງອີງ Book 1 §19/§34/§41 ແລະ STEP 15.

1. ຄວາມຕ້ອງການ (Prerequisites)

- Google Workspace account (ພາຍໃນ Domain — §13)
- ສື່ສ້າງ Google Drive folder ແລະ Spreadsheet
- Node.js + clasp (npm i -g @google/clasp)
- ໂຄດ project (ໂຟນເຕີ photoshop-app/)

2. ທັງ Google Drive (Root Folder)

- ສ້າງ folder ຫຼັກ (ຕົວຢ່າງ: [CompanyName]_DataArchive) → ຄັດລອກ Folder ID ຈາກ URL.
- ensureBaseFolders() ຈະສ້າງ
Photos/Thumbnails/Documents/Temp/RecycleBin/Archive/Backups/System ໃຫ້.

3. Apps Script + clasp

```
npm install -g @google/clasp
clasp login
# ໃນໂຟນເຕີ photoshop-app:
clasp create --type webapp --title "PhotoShop v1.0" # ຫຼື clasp clone <SCRIPT_ID>
clasp push
```

4. Script Properties (§13/§44)

Key	ຄຳ
ROOT_FOLDER_ID	ID ຂອງ Root Folder (Drive)
SHEET_ID	(ທາງເລືອກ) ID Spreadsheet — ຖ້າວ່າ createDatabase ຈະສ້າງ
NOTIFY_TELEGRAM_TOKEN	(ທາງເລືອກ) bot token — ບໍ່ commit

5. ເລີມຖານຂໍ້ມູນ (run ໃນ editor)

```
createDatabase(); // ສ້າງ Sheet ຄົບ
seedDatabase(); // Admin ເລີມຕົ້ນ
validateSchema(); // ຄາດ {ok:true}
reindexAll(); // index ຂໍ້ມູນ (STEP 11)
ensureBaseFolders();// Drive folders
createCleanupTriggers(); // (ທາງເລືອກ) cleanup ບໍ່ມີບັນຫາ
```

6. Deploy Web App (§19)

- Apps Script editor → Deploy → New deployment → Web app
- Execute as: User accessing
- Who has access: ພາຍໃນ Domain (§13) — ຫ້າມ Anyone
- ຄັດລອກ Web App URL ໃຫ້ຜູ້ໃຊ້

7. Backup & Restore (§41)

- Backup: Settings > Backup > "Backup now" (ສໍາເນົາ DB ໄປ Backups folder).
- Restore: Settings > Backup > Restore (ສະຫຼັບ SHEET_ID ໄປ backup — reversible).
- Maintenance: Settings > Backup > "Run maintenance" (cleanup Temp/RecycleBin).

8. Troubleshooting (Error Codes §54)

Code	ວິທີແກ້
E003 Drive Quota	ຍ້າຍ/ລຶບໄຟລ໌ເກົ່າ, ກວດ quota
E004 Folder/Sheet Missing	ກວດ ROOT_FOLDER_ID/SHEET_ID
E005 Permission	ກວດ Role (Settings > Users) & Sharing
E006/E007	ນາມສະກຸນ/ຂະໜາດໄຟລ໌